Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2001-II-Q05] | Part A (Q1-Q5)

Find the largest n such that every ten-element subset of $\{4, 5, \ldots, n\}$ contains three distinct numbers forming a nondegenerate triangle.

求最大 n，使 $\{4, 5, \ldots, n\}$ 的每个十元素子集均含三个不同数可作为非退化三角形边长。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2004-I-Q05] | Part A (Q1-Q5)

Alpha scores $160/300$ on day one and $140/200$ on day two. Beta also attempts 500 points in total, attempts a different first-day amount from 300, earns positive integer scores each day, and has a strictly smaller daily success ratio than Alpha on both days. If Beta's greatest possible total success ratio is m/n reduced, find $m+n$.

Alpha 两天分别得分/作答分值为 $160/300, 140/200$。Beta 两天共作答五百分，首日作答分值不为三百，每天得分为正整数，且两日成功率均严格低于 Alpha 对应日。若 Beta 最大总成功率约分为 m/n，求 $m+n$。

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Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2006-I-Q02] | Part A (Q1-Q5)

Among all ninety-element subsets of $\{1, \ldots, 100\}$, how many distinct element sums occur?

在 $\{1, \ldots, 100\}$ 的所有九十元素子集中，元素和有多少种不同值?

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Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2007-II-Q08] | Part B (Q6-Q10)

Full lines parallel to the sides of a 4×5 paper rectangle have total drawn length 2007. A basic rectangle is bounded by adjacent drawn lines in each direction; paper edges do not count unless drawn. Maximize the basic-rectangle count N and give $N \bmod 1000$.

在 4×5 纸上画与边平行的完整横纵线，总长 2007。基本矩形由两方向各相邻的画线围成，未画出的纸边不算。求最大基本矩形数 N 除以一千的余数。

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Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2022-I-Q07] | Part B (Q6-Q10)

Assign the digits $1, \ldots, 9$ bijectively to $a, b, c, d, e, f, g, h, i$. Minimize the positive value $(abc - def)/(ghi)$, where products are meant. If the minimum is m/n in lowest terms, compute $m + n$.

将 1 至 9 九个数字各用一次分配给 $a, b, c, d, e, f, g, h, i$。式 $(abc - def)/(ghi)$ 中相邻字母表示乘积。求最小正值的最简分数 m/n，并计算 $m + n$。

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Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2025-II-Q08] | Part B (Q6-Q10)

Coins of values $1, 10, 25$ are unlimited. For how many target amounts $1 \le N \le 1000$ does repeatedly taking the largest affordable coin use the minimum possible number of coins?

有无限枚面值 $1, 10, 25$ 的硬币。对 $1 \le N \le 1000$，每次取不超过剩余金额的最大面值硬币。求这种贪心算法能得到最少枚数的金额个数。

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Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2003-II-Q12] | Part C (Q11-Q15)

Each committee member votes for one of 27 candidates. For every candidate, the exact percentage of total votes is at least one less than that candidate's number of votes. Find the smallest possible committee size.

委员会每人给二十七名候选人之一投一票。对每名候选人，其精确得票百分数须比得票数至少小一。求委员会最小可能人数。

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Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2009-II-Q12] | Part C (Q11-Q15)

Choose disjoint pairs of distinct elements from $\{1, \ldots, 2009\}$, requiring all pair sums distinct and at most 2009. Maximize the number k of pairs.

从一至 2009 中选互不重用元素的数对，每对两数不同，所有对和互异且不超过 2009。求最多对数 k。

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Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2022-II-Q14] | Part C (Q11-Q15)

For positive integers $a < b < c$, a stamp collection must include each of these three denominations and must have subcollections totaling every integer from 1 through 1000 cents. Let $f(a, b, c)$ be the least number of stamps in such a collection, when possible. Find the sum of the three smallest c for which some a, b satisfy $f(a, b, c) = 97$.

邮票面值为正整数 $a < b < c$ 分，要求三种面值各至少一张，且用邮票子集能凑出 1 至 1000 的每个整数金额。若能实现，令最少张数为 $f(a, b, c)$。求使某些 a, b 满足 $f(a, b, c) = 97$ 的最小三个 c 之和。

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Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2026-I-Q11] | Part C (Q11-Q15)

Place $1, \ldots, 64$ bijectively in an 8×8 grid. Maximize the sum M of absolute differences over all horizontally or vertically adjacent pairs, and give this maximum modulo 1000.

将 1 至 64 各放一次于 8×8 方格，令 M 为所有水平、竖直相邻格数字差绝对值之和。求最大 M 除以 1000 的余数。

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